INCH-POUND
MIL-PRF-3098/105F
27 August 1997
SUPERSEDING
MIL-C-3098/105E
5 May 1993

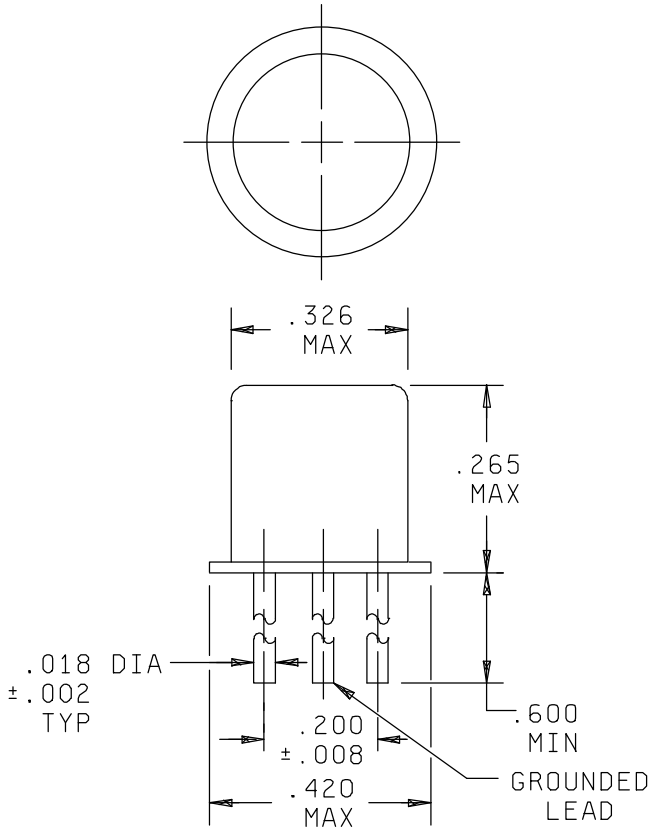
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR103/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 17 MHz to 61 MHz; third mechanical overtone; noncontrolled; series resonance.



Inches	mm
.002	.05
.008	.20
.018	.46
.200	5.08
.265	6.73
.326	8.28
.420	10.67
.600	15.24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR103/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:51Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 17 MHz to 61 MHz, inclusive.

Frequency tolerances: ±25 ppm.

Equivalent resistance: 50 ohms, maximum.

Mode of oscillation: Third overtone.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Capacitance, shunt: 7 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent-resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent-resistance change permitted: ±10 percent.

Vibration: Method 204, MIL-STD-202, test condition B.

Frequency change permitted: ±5 ppm.

Equivalent-resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±2.5 ppm.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force -85

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 17, 19, 99

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-0697-69)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:51Z
Check the source to verify that this is the current version before use.